

CHARTERED INSTITUTE OF TECHNOLOGY, ABU ROAD
B.Tech. 1st Sem, - Internal Examination, 2024-25
EME

Roll No. CSE-53

Suraj Singh

Year/ Sem:- 1Year / 1st Sem
Faculty Name:- MOSIN SIR

Time :- 10.00 AM 11.30 AM

DATE: 18.11.2024
Maximum Marks: 40

Instruction to Candidate:-

Part -A: - Short Answer question 5 x 2 marks= 10 marks. All **Five** questions are **compulsory**.

Part- B: - Analytical/ Problem solving question 2 x 8 marks = 16 marks. Candidate has to answer **Two** question out of **four**.

Part-C: - Descriptive question 1 x 14 marks = 14 marks. Candidate has to answer **one** question out of **three**.

Part -A

- Q1.** Define a thermodynamic system and classify the different types.
- Q2.** Explain the Zeroth Law of Thermodynamics and its practical application.
- Q3.** What is internal energy, and why is it significant in thermodynamics?
- Q4.** Differentiate between heat and work with appropriate examples.
- Q5.** State and explain the First Law of Thermodynamics for a closed system

Part-B

- Q1. Theory Question:** State and explain the Second Law of Thermodynamics. Discuss its implications in real-life applications.
- Q2. Theory Question:** Describe the First Law of Thermodynamics for a control volume and provide examples of its application.
- Q3. Numerical Question:** A gas undergoes an isobaric process where 100 kJ of heat is added. The initial volume is 0.5 m^3 , and the final volume is 1.0 m^3 . Calculate the work done.
- Q4.** An ideal gas at 320 K, 320K expands isothermally from an initial volume of 0.4 m^3 to 1.5 m^3 . The initial pressure is 180kPa. **Find:**

- III. The work done by the gas.
- IV. The heat transferred during the process.